Table 3. **LPDMA1 and LPDMA2 autonomous mode and wake-up in low-power modes**

Feature	Low-power modes
Wake-up	LPDMA1/2 in Sleep mode

3.13 Interrupts and events

3.13.1 Nested vectored interrupt controller (NVIC)

- 88 maskable interrupt channels (not including the 16 Cortex®-M33 with FPU interrupt lines)
- 16 programmable priority levels (4 bits of interrupt priority used)
- Low-latency exception and interrupt handling
- Power management control
- Implementation of system control registers

The NVIC and the processor core interface are closely coupled, enabling low-latency interrupt processing and efficient processing of late-arriving interrupts. All interrupts, including the core exceptions, are managed by the NVIC.

3.13.2 Extended interrupt and event controller (EXTI)

The extended interrupts and event controller (EXTI) manages the individual CPU and system wake-up through configurable and direct event inputs. It provides wake-up requests to the power control and generates an interrupt request to the CPU NVIC and events to the CPU event input. For the CPU, an additional event generation block (EVG) is needed to generate the CPU event signal.

The EXTI wake-up requests allow the system to wake up from Stop modes.

The interrupt request and event request generation can also be used in Run modes.

The EXTI also includes the EXTI mux I/O port selection.

The EXTI main features are the following:

- 37 input events are supported.
- All event inputs allow the possibility to wake up the system.
- Events that do not have an associated wake-up flag in the peripheral have a flag in the EXTI and generate an interrupt to the CPU from the EXTI.
- Events can be used to generate a CPU wake-up event.

The asynchronous event inputs are classified into two groups:

- Configurable events (signals from I/Os or peripherals able to generate a pulse), with the following features:
  - Selectable active trigger edge
  - Interrupt pending status register bits independent for the rising and falling edge
  - Individual interrupt and event generation mask, used for conditioning the CPU wake-up, interrupt, and event generation
  - Software trigger possibility
  - EXTI I/O port selection
- Direct events (interrupt and wake-up sources from peripherals having an associated flag which requires to be cleared in the peripheral), with the following features:
  - Fixed rising edge active trigger
  - No interrupt pending status register bit in the EXTI (the interrupt pending status flag is provided by the peripheral generating the event)
  - Individual interrupt and event generation mask, used to condition the CPU wake-up and event generation
  - No software trigger possibility